

AI Social Media Auditor

Python Practical Submission

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1. Program Description

This program has two main parts. First, it counts the frequency of words from a paragraph. Before counting, the words are converted to lowercase and punctuation is removed. This makes words such as AI, ai, and ai, count as the same word.

Second, the program compares followers from two social media platforms using Python sets. Set intersection is used to find users present on both platforms, set difference is used to find users present only on one platform, and symmetric difference is used to find users present on exactly one platform.

2. Python Code

```
1
2 # AI Social Media Auditor
3
4 """
5 AI Social Media Auditor
6
7 This program has two small parts:
8 1. Count words from a paragraph.
9 2. Compare followers from two social media platforms.
10
11 Before counting, I change words to lowercase and remove
12     punctuation.
13 This makes words like "Python", "python" and "python," count as
14     one word.
15
16 For followers:
17 & means users present on both platforms
18 - means users present only on one particular platform
19 ^ means users present on exactly one of the two platforms
20 """
21
22 import string
23
24 # ----- TASK 1: WORD COUNTER -----
25
26 text = ""
```

```

26 AI is changing the world.ai is powering new tools,and AI is
    helping
27 researches learn faster.The world loves loves AI ,and the world
28 depends on it more each year.
29 """
30
31 word_counts = {}
32
33 for raw_word in text.split():
34
35     # make every word lowercase and remove punctuation
36     cleaned_word = raw_word.lower().strip(string.punctuation)
37
38     if not cleaned_word:
39         continue
40
41     word_counts.setdefault(cleaned_word, 0)
42
43     word_counts[cleaned_word] += 1
44
45
46 print("==== WORD FREQUENCY ====")
47
48 for word, count in word_counts.items():
49     print(word, ":", count)
50
51
52 # ----- TASK 2: FOLLOWER COMPARISON -----
53
54 platform_a_followers = [
55     "user101",
56     "user102",
57     "user103",
58     "user104",
59     "user105"
60 ]
61
62 platform_b_followers = [
63     "user103",
64     "user104",
65     "user106",
66     "user107"
67 ]
68
69
70 # Convert lists into sets
71 set_a = set(platform_a_followers)
72 set_b = set(platform_b_followers)
73
74
75 # Users present on both platforms

```

```

76 both_platforms = set_a & set_b
77
78
79 # Users only on Platform A
80 only_a = set_a - set_b
81
82
83 # Users only on Platform B
84 only_b = set_b - set_a
85
86
87 # Users present on exactly one platform
88 either_only = set_a ^ set_b
89
90
91 print("\n==== FOLLOWER REPORT =====")
92
93 print("Users on both platforms:", both_platforms)
94
95 print("Only Platform A:", only_a)
96
97 print("Only Platform B:", only_b)
98
99 print("Only one platform:", either_only)

```

3. Program Output

The output obtained after running the Python program is shown below.

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win
32
Enter "help" below or click "Help" above for more information.
->
= RESTART: C:/Users/Sneha Gautam/Downloads/COLLEGE/AI/Assignment 8_lab/AI_Social_Medi
_Auditor.py
===== WORD FREQUENCY =====
ai : 3
is : 3
changing : 1
the : 2
world.ai : 1
powering : 1
new : 1
tools,and : 1
helping : 1
researches : 1
learn : 1
faster.the : 1
world : 2
loves : 2
and : 1
depends : 1
on : 1
it : 1
more : 1
each : 1
year : 1

===== FOLLOWER REPORT =====
Users on both platforms: {'user104', 'user103'}
Only Platform A: {'user105', 'user102', 'user101'}
Only Platform B: {'user107', 'user106'}
Only one platform: {'user107', 'user106', 'user102', 'user101', 'user105'}
-> |
```

Figure 1: Output of AI Social Media Auditor

4. Concepts Used

- **String Processing:** Words are converted into lowercase and punctuation is removed.
- **Dictionary:** The dictionary `word_counts` stores words and their frequencies.
- **List:** Follower usernames are initially stored in lists.
- **Set:** Lists are converted into sets for easy comparison.
- **Intersection (&):** Finds users present on both platforms.
- **Difference (-):** Finds users present only on one platform.
- **Symmetric Difference (^):** Finds users present on exactly one platform.

5. Conclusion

The AI Social Media Auditor demonstrates the use of basic Python data structures and operations. It counts word frequencies using a dictionary and compares social media followers using sets. The program also demonstrates loops, string processing, list-to-set conversion, and different set operations.